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M.Tech.
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. CSE	Indian Institute of Technology Indore	Current	2026-Present
B.Tech. CSE	University of Calcutta	9.0/10.0	2022-26
Senior Secondary	CBSE Board	84.8%	2021
Secondary	CBSE Board	94.6%	2019

Experience

- **Extended Dynamic Systems** *Nov 2025 - May 2026*
  Software Developer Intern Kolkata, India
  – Engineered a distributed microservices architecture spanning 4+ Dockerized Go services and implemented fine-grained RBAC across 6 user and administrative roles.
  – Developed 30+ REST APIs and 15+ gRPC services integrating AWS Cognito passwordless authentication, S3 object storage, and OpenTelemetry distributed tracing.
  – Architected asynchronous event-driven workflows using AWS SQS, provisioned infrastructure with Terraform, and automated CI/CD pipelines using GitHub Actions.

Projects

- **PenguinDB** *2025 - Present*
  Custom SQL Database Github
  – Contributed to the SQL layer of a custom database supporting SQL queries over an underlying key-value storage engine.
  – Implemented the SQL lexer, parser, and AST nodes, establishing the foundation for query planning and execution.
  – Worked with database internals including WAL, SSTables, LSM Trees, manifests, Bloom filters, storage engine, catalog, and encoding.

- **Cravyn - Food Delivery App** *Sep. 2024 - Jan. 2025*
  Full Stack Developer Website
  – Built a monolithic backend REST API using Express.js and Bun to serve a responsive Next.js 15 frontend.
  – Designed and implemented a custom JWT-based authentication system using HttpOnly cookies for secure user sessions.
  – Designed a relational PostgreSQL schema for customers, restaurants, food items, carts, addresses, and order processing.

Technical Skills

- **Programming**: C, C++, Python, JavaScript, TypeScript, Go
- **Frameworks & Libraries**: React, Express.js, Next.js, Hono.js, Tailwind CSS
- **Databases**: PostgreSQL, MySQL, SQLite, MongoDB, Redis, Elasticsearch
- **Tools & Platforms**: Docker, Kubernetes, Git, GitHub Actions, AWS, Terraform
- **Core Concepts**: Data Structures & Algorithms, REST APIs, gRPC, Microservices, Database Systems

Key courses taken

- **Mathematics**: Linear Algebra, Probability
- **Computer Science**: Operating Systems, Compiler Design

Achievements

- **1st Place**,Status Code 2 Hackathon – Quantitative Finance and RISE Tracks
- **AIR 1951**,GATE 2025 – Computer Science *2025*
- **Encouragement Award** ,It's Alive – AI NPC Game Jam
- **LeetCode** ,Rating: 1918, Top 4.17%
- **Codeforces** ,Rating: 1100+